

Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects*

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Abstract

To estimate the dynamic effects of an absorbing treatment, researchers often use two-way fixed effects regressions that include leads and lags of the treatment. We show that in settings with variation in treatment timing across units, the coefficient on a given lead or lag can be contaminated by effects from other periods, and apparent pretrends can arise solely from treatment effects heterogeneity. We propose an alternative estimator that is free of contamination, and illustrate the relative shortcomings of two-way fixed effects regressions with leads and lags through an empirical application.

Keywords: difference-in-differences, two-way fixed effects, pretrend test

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1.1 Related Literature

This paper makes two main contributions within an active literature on the causal interpretations of two-way fixed effects models in settings with staggered treatment adoption (Athey and Imbens, 2018; Borusyak and Jaravel, 2017; Callaway and Sant’Anna, 2020a; de Chaisemartin and D’Haultfœuille, 2020; Goodman-Bacon, 2018). Our paper is also related to the traditional literature analyzing non-separable panel and treatment effects models e.g. Heckman et al., 1998, 1997; Blundell et al., 2004; Abadie, 2005; Chernozhukov et al., 2013.

The first main contribution of our paper is to interpret estimates from two-way fixed effects specifications when researchers include “dynamic” indicators for time relative to treatment and when treatment effects are heterogeneous across adoption cohorts. We derive our results for a general class of two-way fixed effects specifications where “dynamic” indicators can be flexibly specified as single relative periods ℓ or sets of relative periods g (thus also capturing any “static” specification where all post-treatment indicators are collected in a single set). This class of specifications encompasses all specifications addressed by Athey and Imbens (2018), Borusyak and Jaravel (2017), Callaway and Sant’Anna (2020a), de Chaisemartin and D’Haultfœuille (2020) and Goodman-Bacon (2018).

As a building block for the causal interpretation of estimates, we define $CATT_{e,\ell}$, the cohort average treatment effects on the treated as the cohort-specific average difference in outcomes relative to never being treated. Our choice of a “building block” is governed by the counterfactual and the type of heterogeneity of interest. This object coincides with the “group-time average treatment effect” studied by Callaway and Sant’Anna (2020a) and is more granular than the building block used by Goodman-Bacon (2018) that is an average of $CATT_{e,\ell}$ over some relative period range. Athey and Imbens (2018) consider an alternate counterfactual to never being treated: being treated at a different time. Borusyak and Jaravel (2017) implicitly assume away heterogeneity across cohorts within a relative period, so their building block reduces to ATT_ℓ . de Chaisemartin and D’Haultfœuille (2020) allow for heterogeneous treatment paths within a cohort over time across “groups”, thus their building block is at the group level. We defer a discussion of assumptions underlying the causal interpretation of these building blocks to Section 2.

The second main contribution of our paper is to propose a simple regression-based alternative estimation strategy that produces a more sensible estimand than conventional two-way fixed effects models under heterogeneous treatment effects. Our procedure is most similar to Callaway and Sant’Anna (2020a), but has the

the closed-form expressions for the weights using the classical “omitted variables bias formula” in Section 3.6. We defer the formal derivation to the Appendix. Here we mention the following four properties of the weights $\omega_{e,\ell}^g$.

- For relative periods of μ_g ’s own bin i.e. $\ell \in g$, their associated weights as displayed in (9) sum to one $\sum_{\ell \in g} \sum_e \omega_{e,\ell}^g = 1$.
- For relative periods belonging to some other bin included in (4) i.e. $\ell \in g'$ for $g' \neq g$ and $g' \in \mathcal{G}$, their associated weights as displayed in (10) sum to zero $\sum_{\ell \in g'} \sum_e \omega_{e,\ell}^g = 0$ for each bin g' .
- For relative periods not included in $\mathcal{G}$, their associated weights as displayed in (11) sum to negative one $\sum_{\ell \in g^{excl}} \sum_e \omega_{e,\ell}^g = -1$.
- If there are never-treated units i.e. $\infty \in \text{supp}(E_i)$, we have $\omega_{\infty,\ell}^g = 0$ for all g and ℓ .

We can easily estimate the weights $\omega_{e,\ell}^g$ for any given specification of $\mathcal{G}$ using the following auxiliary regression:

$$D_{i,t}^\ell \cdot \mathbf{1}\{E_i = e\} = \alpha_i + \lambda_t + \sum_{g \in \mathcal{G}} \omega_{e,\ell}^g \mathbf{1}\{t - E_i \in g\} + v_{i,t} \quad (12)$$

which regresses $D_{i,t}^\ell \cdot \mathbf{1}\{E_i = e\}$ on all bin indicators included in regression (4) and two-way fixed effects.

All of the above properties can be extended to a case where covariates are added to regression (4) by partialling out the covariates before proceeding. In other words, the terms in parentheses in (9), (10) and (11) would be replaced by terms for which the covariates are partialled out. The weights can be estimated by controlling for covariates in regression (12) the same way as they are controlled for in the original regression.

However, covariates complicate the interpretations of μ_g in terms of $CATT_{e,\ell}$ as we describe below in Proposition 2-4. Depending on how covariates are controlled for in regression (4), we may need an additional assumption that the counterfactual trends are linear in the time-varying covariates $X_{i,t}$. We leave a full investigation of the introduction of covariates for future work.

3.3 Interpreting the coefficients under parallel trends assumption only

Proposition 2. *Under Assumption 1 (parallel trends) only, the population regression coefficient on the indicator for relative period bin g is a linear combination of $CATT_{e,\ell \in g}$ as well as $CATT_{e,\ell'}$ from other*